

Usman Aslam

Full-stack developer with ML experience | C# | Python | Angular | Node | React | DevOps | Azure | Jenkins | Docker | Kubernetes | Data Science | EF | SQL | No SQL | MongoDB | .NET | REST API

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Professional Summary

Senior Software Engineer with 10+ Years of Experience in Software Development and Architecture, experience specializing in C#, Java, and Python. Expertise in collaborating with cross-functional teams to design and deliver scalable, high-performance software solutions. Strong background in API design, microservices architecture, and mentoring junior engineers to enhance technical proficiency. Proven ability to evaluate and implement modern database solutions tailored to project needs. Committed to elevating code quality through thorough reviews, significantly reducing defects. Continuously staying updated on emerging technologies and fostering the adoption of best practices. Highly effective communicator, working closely with stakeholders to ensure timely and successful software product launches.

Skills

C# .NET and .NET Core • Python • Blazor • ASP.NET MVC • TypeScript • jQuery • JavaScript • HTML/CSS • React • Angular 8+ • VUE • Sklearn & Pytorch • Web & API Development • Docker • MongoDB • Flask • Prompt Engineering • LLM (OpenAI, Llama, Gemini) • Microsoft Azure • Microservices • MS SQL Server • SOAP • DevOps • RAG System using LLM • Git • Azure DevOps • Jenkins • CI/CD • Cloud Computing Azure Services • Coding and debugging technique • Kubernetes • Model Training • Fine Tuning • Foursquare API • API Gateway • GraphQL • Langchain & Llamaindex • Tensorflow & Keras • Generative Ai • Gathering System Requirement • Change Request Management • Conduct Scrum on the bases of Scrum Knowledge • Factory Pattern • Dependency Injection Pattern • Command Pattern • Repository Pattern • CQRS • Kabana & Elastic Search • Vector Database FISS, qDrant • LINQ • Entity framework • AWS Cloud service • Lambda, SNS, API Gateway, DynamoDB • RabbitMQ

Work Experience

Dubai Land Department - ERES EmaraTech, Dubai • Dubai

Dec 2021 - Jan 2025

Sr. Software Engineer

- **End-to-End Software Development:** Designed, developed, tested, and deployed software solutions using technologies such as **C#, Python, and .NET Core**, managing the entire product lifecycle from initial concept through to production launch.
- Proficient in **backend technologies** such as **Node.js, Fast API** and **Flask**, with the ability to build RESTful APIs and integrate with frontend systems.
- Collaborated with **domain experts, UX/UI designers, and Product Managers** to translate organizational needs into actionable technical solutions, ensuring alignment between business goals and development efforts.
- Led architectural and design discussions on key platform features, providing **technical expertise in microservices architecture** and **cloud-based solutions (AWS, Azure)**, ensuring performance optimization and scalability.
- Spearheaded the development, fine-tuning, and deployment of **Generative LLMs** and **RAG (Retrieval-Augmented Generation) models** using **TensorFlow(Keras)** and **PyTorch**, integrating **vector databases** to power **conversational AI products**. Demonstrated expertise in **NLP**, and AI workflows, optimizing models for real-time performance, and ensuring high-quality, contextually relevant outputs for a variety of applications.
- **Prototyped and conducted live experiments** to test new features and gather feedback, ensuring that development efforts were aligned with end-user needs and system requirements.

- **Cloud & Containerization:** Implemented containerization strategies using Docker and orchestrated services using Kubernetes to ensure seamless deployment, scaling, and management of applications in cloud environments.
- **Designed and developed applications using a range of databases,** including **NoSQL** (MongoDB) and **relational databases** (SQL Server, PostgreSQL, Oracle), ensuring efficient data modeling, optimized queries, and seamless integration to meet application performance and scalability requirements.
- Conducted regular **code reviews** and maintained high standards of **code quality**, ensuring the team followed best practices in **C#, Python**, and Restful **API design**.
- Drove collaboration across **Agile teams** to ensure successful feature development and iterative releases, enabling faster time-to-market for new products and features.
- **Agile Development & CI/CD:** Drove Agile development practices within cross-functional teams to ensure iterative, high-quality releases. Implemented CI/CD pipelines to automate deployment and accelerate the time-to-market for new features.
- **Architectural Expertise:** Led architectural discussions on key platform features, focusing on microservices-based architectures, cloud-based solutions (AWS, Azure), and performance optimization to ensure scalability, security, and reliability.
- **Leadership & Ownership:** Took end-to-end ownership of features, leading cross-functional teams to ensure alignment with business objectives and adherence to technical standards throughout the software development lifecycle.

Abu Dhabi Chamber of Commerce, Abu Dhabi • Abu Dhabi

Jul 2015 - Oct 2021

Software Engineer

- **Developed and maintained complex .NET applications** using C# and ASP.NET, ensuring scalability, performance, and security.
- **Designed and implemented RESTful APIs** for integration with external systems, third-party services, and CRM platforms, ensuring seamless data exchange.
- **Led the architecture and development** of high-performance, distributed systems, utilizing microservices and service-oriented architectures (SOA).
- **Optimized SQL Server databases** by improving queries, indexing strategies, and ensuring efficient data retrieval and storage.
- **Collaborated with front-end teams** to integrate React/Angular portals with backend services, ensuring smooth, responsive user interfaces.
- **Implemented and maintained CI/CD pipelines** using Azure DevOps, TFS, and Git for automated testing, build, and deployment processes.
- **Led code reviews and mentoring sessions**, promoting best practices in .NET development, design patterns, and code quality.
- **Managed and optimized legacy .NET applications**, refactoring outdated code, improving performance, and reducing technical debt.
- **Conducted performance tuning and debugging**, identifying bottlenecks, optimizing server response times, and troubleshooting complex production issues.
- **Collaborated with product managers, UX/UI teams, and business stakeholders** to gather requirements and deliver tailored solutions on time and within scope.
- **Led migration projects**, transitioning monolithic systems to microservices or cloud-native architectures, ensuring smooth data migration and minimal downtime.
- **Monitored application performance and reliability**, using tools like Application Insights, ELK stack, and SQL Profiler to ensure high availability.
- **Provided technical leadership and strategic direction** for .NET development teams, ensuring alignment with business goals and best practices.
- **Delivered client-facing solutions**, tailoring .NET-based software to meet specific customer needs and business requirements.

- **Demonstrated ability to work independently in a consultative, client-facing role**, providing tailored solutions and strategic advice to meet client needs and drive business outcomes.
- Defined and documented **business requirements** for the **CRM system**, collaborating with **domain experts** and **product managers** to design solutions that aligned with organizational goals.
- Improved **code quality** by revising complex programs, documenting code for readability, and optimizing software to meet **performance standards**.

Ovis Global, Lahore Pakistan • Lahore

Jan 2014 - Apr 2015

Software Engineer

- **Developed solutions by evaluating alternative workflows**, prepared technical specifications, and collaborated closely with teams to determine the best approach using **.NET (C#, ASP.NET Core)**, **SQL Server**, and **RESTful APIs** for backend services.
- **Defined solution specifications** and coordinated development efforts with programmers using **Azure DevOps** and **Git** for version control, ensuring timely and high-quality delivery.
- **Collaborated with clients** to understand their business challenges and provided tailored solutions using **.NET MVC** and **Entity Framework** for data management and business logic.
- **Designed and built a custom E-commerce CMS** for clients, enabling them to manage and update their websites seamlessly, utilizing **.NET Core**, **React** for front-end interfaces, and **SQL Server** for database management.
- **Managed Jira-based development** for issue tracking, sprint planning, and project management, ensuring efficient workflow and support during the development cycle.
- **Developed a custom CRM system** for generating client reports, sending notifications, and managing user data, using **C#**, **ASP.NET MVC**, and **SQL Server**.
- **Integrated user web applications with existing systems**, ensuring seamless data flow using **RESTful APIs** and **Web API** services for backend integration.
- **Translated clients' business requirements** into technical specifications, leveraging **.NET technologies**, **SQL Server**, and **API integration** to develop scalable, efficient solutions.
- **Collaborated with the Quality Assurance (QA) team**, ensuring that solutions were thoroughly tested and delivered bug-free using **NUnit**, **Selenium**, and other automated testing tools.

Projects

FAQ System Using RAG

ERES

I developed the **RAG FAQ Generator**, leveraging the organization's data stored in a **Vector Store Database** to automate the creation of relevant FAQs from documentation. I designed and integrated the system with a local **LLM**, utilizing **Prompt Engineering** to ensure accurate and meaningful question generation. This project streamlined the FAQ creation process, enhanced knowledge sharing across the team, and significantly improved efficiency by reducing manual efforts in content extraction and response generation.

Mollak - Property Management System

Dec 2021 - Present

Dubai Land Department - ERES

The Mollak portal is an online platform designed to manage service charges for jointly-owned properties in Dubai. It helps property owners and management companies track financial transactions, ensure transparency, and maintain accurate accounting of shared expenses.

Key Features:

- **Service Charge Management:** Simplifies the calculation and collection of service charges.
- **Financial Reporting:** Offers detailed financial statements for stakeholders.
- **Transparency:** Allows owners to access important financial information.
- **Regulatory Compliance:** Ensures property management aligns with local regulations.

Overall, Mollak enhances the governance of jointly-owned properties, contributing to a more organized real estate market in Dubai.

Technology Stack:

- **Frontend:** Angular 8 for a responsive UI.
- **Backend:** .NET (C#) for server-side development.
- **Database:** MongoDB for scalable log data storage.
- **Messaging:** RabbitMQ for inter-component communication.
- **Payment Integration:** SMS and WhatsApp gateways for transactions.
- **Containerization:** Kubernetes for application orchestration.
- **Scheduled Tasks:** Cron jobs for automating processes.
- **DevOps:** CI/CD pipelines for efficient updates and maintenance.

Education

BS, Software Engineering <i>University of Engineering & Technology, Taxila, Pakistan</i>	Aug 2009 - Jan 2014
Intermediate <i>BISE Lahore, Pakistan</i>	Jul 2007 - Aug 2009

Certifications and Licenses

Generative AI Project Management Institute • Jun 2024
AZ-204: Developing Solutions for Microsoft Azure Microsoft • 2023
Microsoft Azure (AZ-900) Microsoft • 2021
IBM Applied AI by Coursera Coursera • 2020
IBM Data Science by Coursera Coursera • 2020

Relevant Coursework

Software Architecture • Software Construction • Software Requirement Engineering • Software Quality Assurance • Software Development • Microservices • Generative AI